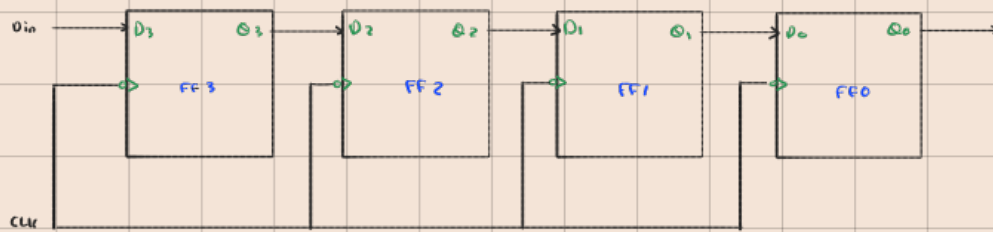


Shift Registers

SISO

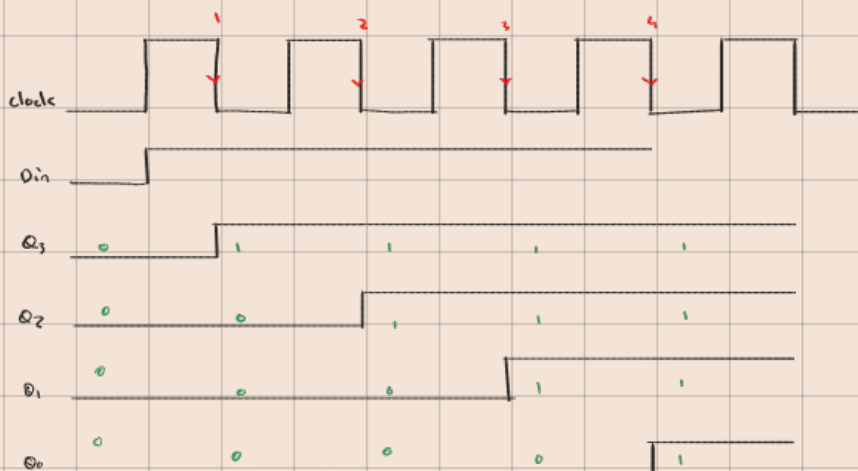
- serial input
- serial output



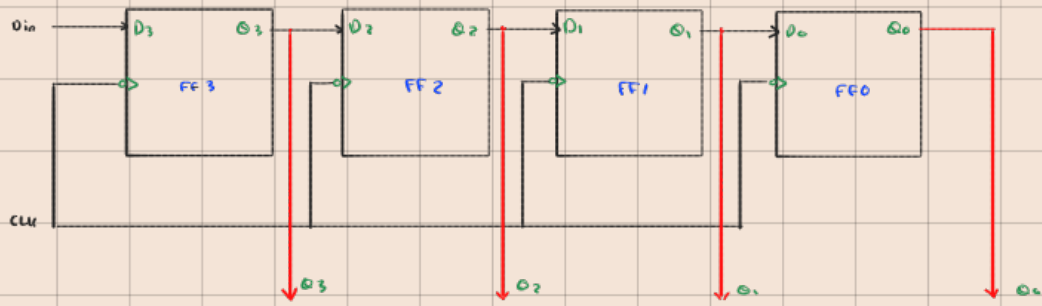
Input: 1111

CLK	Q3	Q2	Q1	Q0
initial	0	0	0	0
	1	0	0	0
	1	1	0	0
	1	1	1	0
	1	1	1	1

wave Diagram



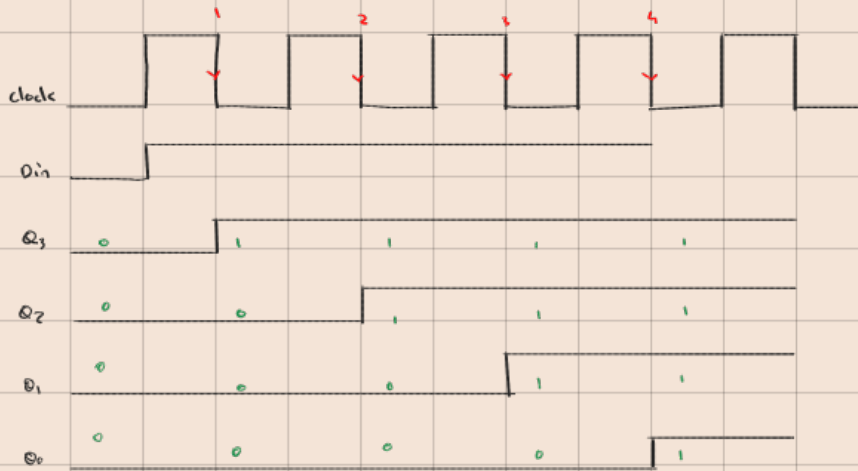
SISO:



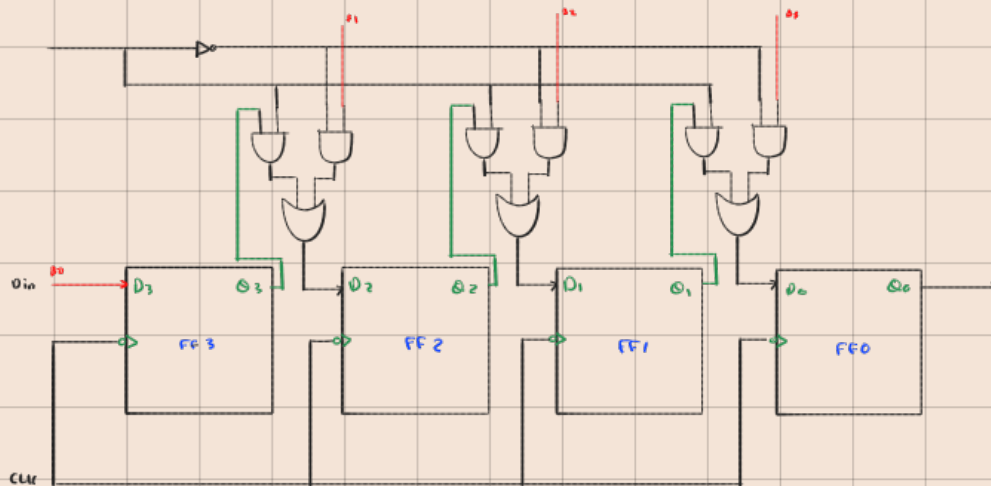
inp: 1111

clk	Q3	Q2	Q1	Q0
initial	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1

wave diagram



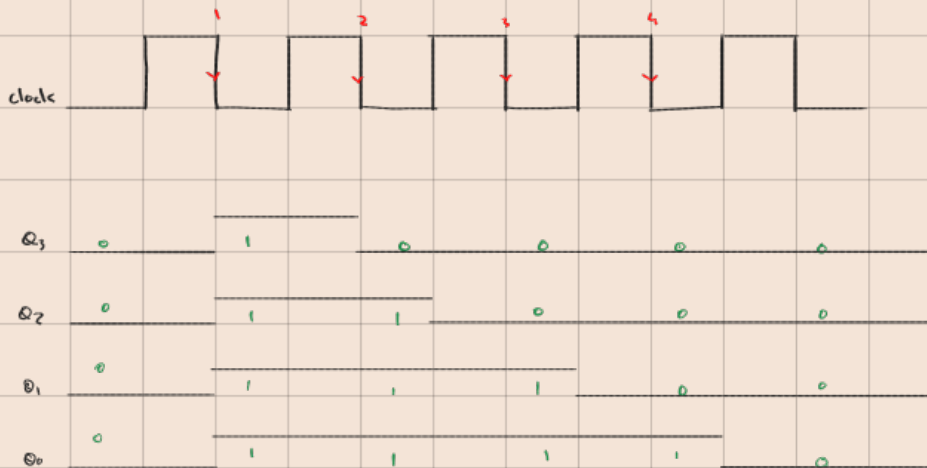
Piso :



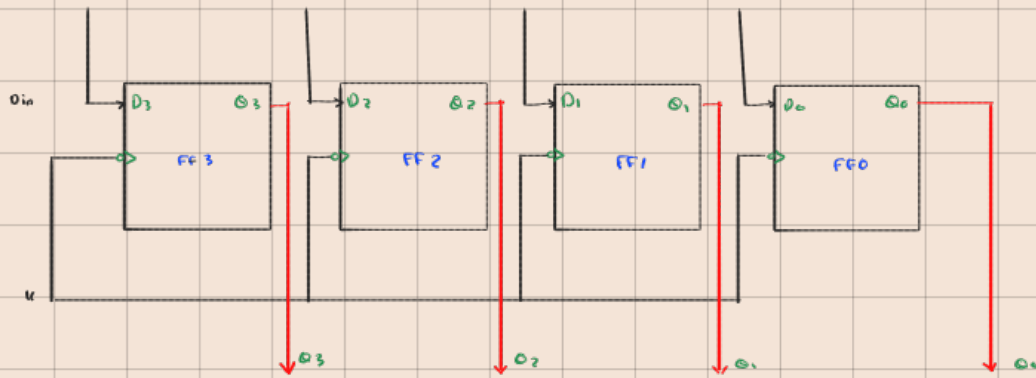
inp: 1111

clk	Q3	Q2	Q1	Q0
initial	0	0	0	0
low mode	1	1	1	1
shift to	0	1	1	1
	0	0	1	1
	0	0	0	1

wave diagram



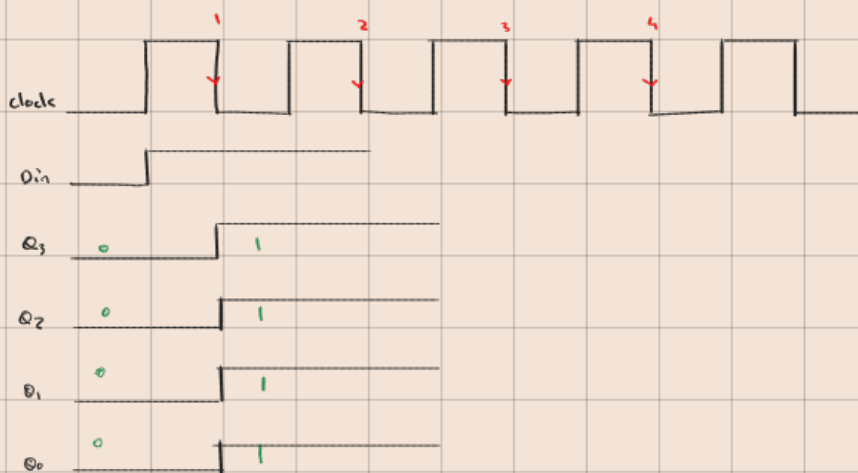
PiPo: (Storage / Buffer Register)



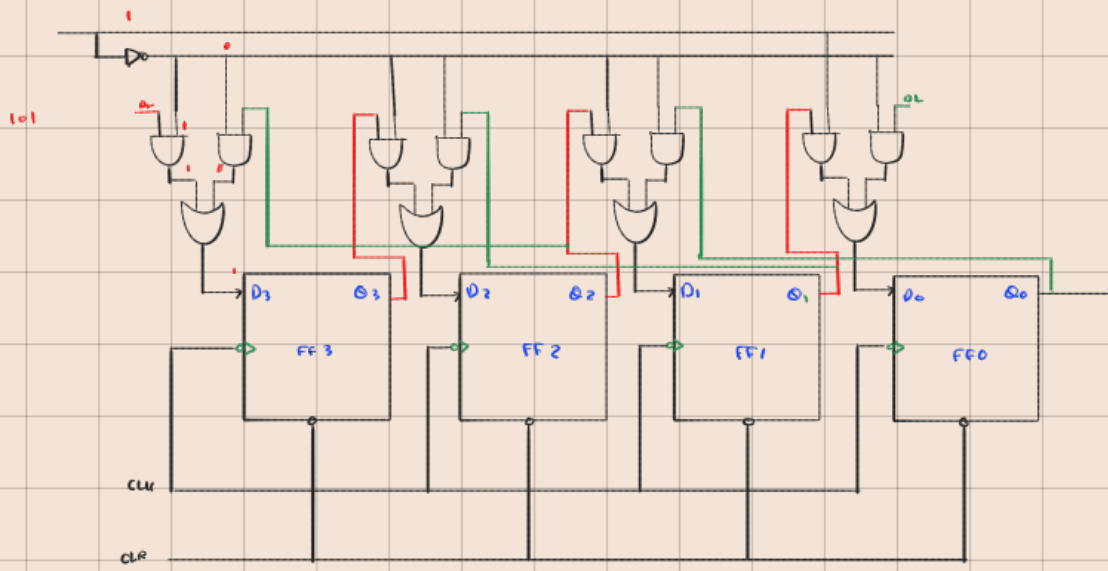
Input: 1111

clk	Q3	Q2	Q1	Q0
initial	0	0	0	0
	1	1	1	1

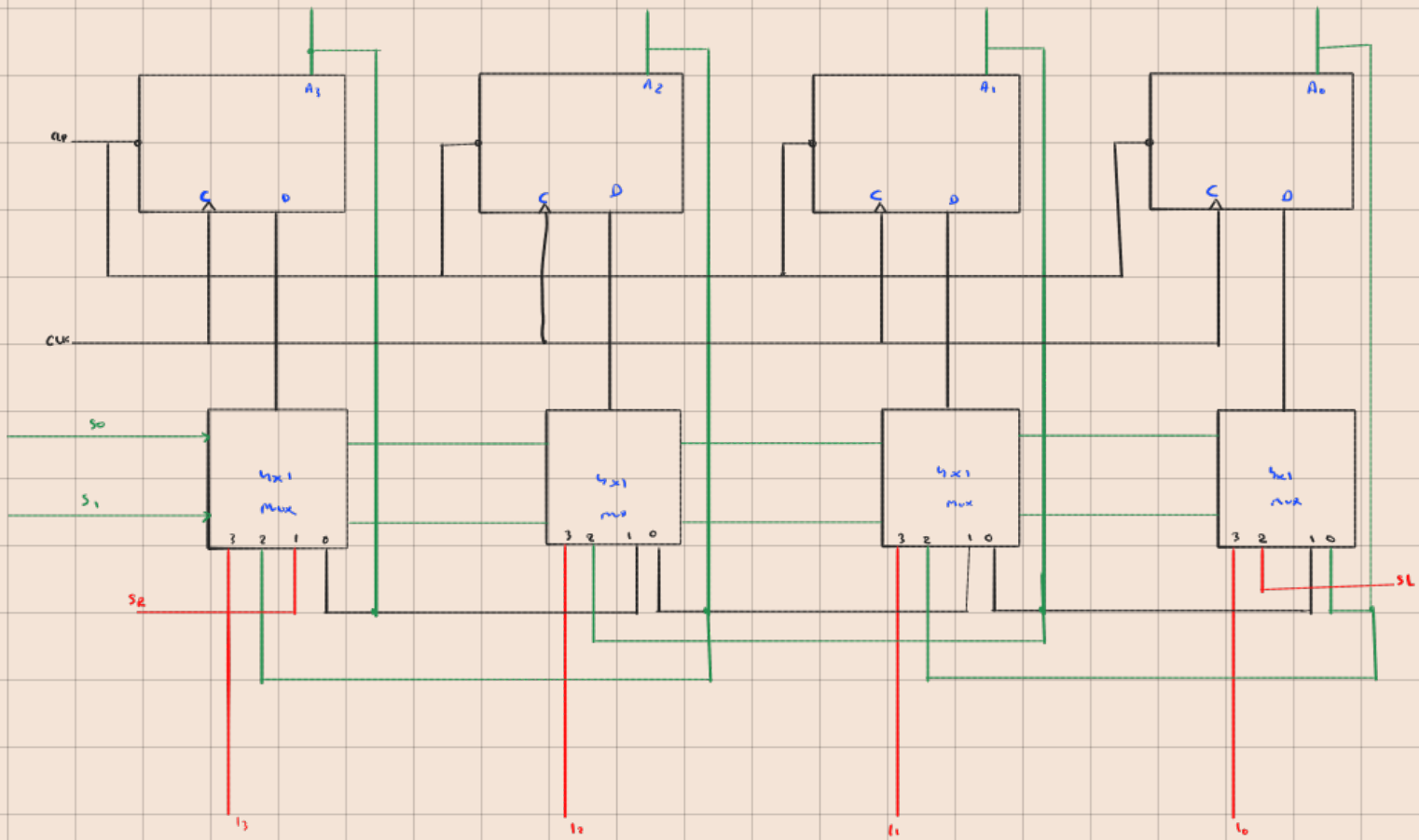
wave Diagram



BiDirectional Register



Universal Register



n

S_0 S_1

operation

0 0

No change

0 1

right shift

1 0

left shift

1 1

parallel loading

Input Q_3, Q_2, Q_1, Q_0
(011)

clk	Q_3	Q_2	Q_1	Q_0	D_{in}
init	0	0	0	0	
	1	0	0	0	1
	0	1	0	0	0
	1	0	1	0	1
	1	1	0	1	1

4 bit sync counter X pdf

	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	1	0	0	0
5	1	0	0	1
6	1	1	0	0
7	1	1	1	1
8	0	0	0	0

wave Diagram

